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Engineering Research Project Part 1

*Reducing Perceptual Lag in Temporal Vision Models for
Autonomous Suburban Driving*

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Abstract

The nUWAY shuttle project at UWA is working toward SAE Level 4 autonomy on the UWA campus. Quirke-Brown’s prior work, conducted on the Eglinton suburban route, established VMamba as the strongest-performing architecture among those tested, achieving 95.77% autonomous driving time against video-based temporal models that all performed worse in deployment [1]. One temporal model was pulled after a single run on safety grounds.

Offline metrics did not predict this failure. All models ran above the required frame rate and achieved comparable training performance to the single-frame baselines. The degradation appears architectural: video models designed for classifying pre-recorded footage aggregate temporal information across a fixed input window, producing outputs that reflect a blended summary of the input history rather than the current scene. Quirke-Brown attributes the observed deployment failures to this perceptual lag and identifies several contributing design choices: bidirectional scanning, fixed window length, uniform temporal weighting, and fixed inter-frame spacing; as candidates for modification [1].

This thesis investigates whether targeted architectural changes can close the gap. A new dataset is collected on the UWA campus route and VMamba is retrained on it to establish a campus-specific baseline. Four modifications are then applied to the temporal architecture: forward-only causal scanning, adaptive frame windowing, recency-biased temporal weighting, and varied inter-frame spacing. The hypothesis is that at least one configuration will match or exceed the campus VMamba baseline by reducing perceptual lag without sacrificing the motion context that temporal representations can provide.

Each modification is implemented in PyTorch and screened offline against training loss, validation loss, and per-behaviour accuracy before real-world testing on the shuttle. Evaluation follows Quirke-Brown’s protocol: autonomous driving time, disengagement and intervention counts, steering smoothness, path deviation, and ride comfort.

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Nomenclature

Latin symbols

$\mathbf{A}, \mathbf{B}, \mathbf{C}$	SSM state, input, and output projection matrices
f_t	Video frame at discrete time step t
$\mathbf{h}_t$	Hidden state vector at time step t
k	Lag offset (number of steps before current frame)
T	Temporal window length (number of input frames)
t	Discrete time step index
w_i	Recency weight assigned to frame i
α_t	Per-frame adaptive calibration weight (TAdaConv)

Greek symbols

Δ	Input-dependent discretisation step size (Mamba Δ parameter)
Δt	Inter-frame time spacing
τ	Recency decay factor

Acronyms

CARLA	Car Learning to Act (open-source driving simulator)
CNN	Convolutional Neural Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
Hz	Hertz (cycles per second; used for camera frame rate)
LTI	Linear Time-Invariant
MAE	Mean Absolute Error
MaIL	Mamba Imitation Learning (robotic manipulation policy)
MTIL	Mamba-based Temporal Imitation Learning
RGB	Red–Green–Blue (colour image representation)
ROS2	Robot Operating System 2
SAE	Society of Automotive Engineers
SS2D	2D Selective Scan (VMamba spatial scan module)
SSM	State Space Model
UWA	University of Western Australia

1 Introduction



Figure 1: The nUWay EasyMile EZ10 shuttle Bus. [2].

Autonomous shuttle systems operating on public roads face a problem that benchmarks do not expose: offline classification performance does not predict real-time control reliability. The nUWay project at the University of Western Australia is developing an EasyMile EZ10 shuttle for SAE Level 4 operation on the UWA campus, an environment with pedestrian crossings, narrow service roads, shared paths, and mixed-use intersections. Unlike controlled motorway settings, this requires continuous low-latency decisions across a wide variety of manoeuvres with no tolerance for reaction delay.

Prior work by Quirke-Brown, conducted on the Eglinton suburban route, established VMamba as the strongest-performing architecture among those tested, achieving 95.77% autonomous driving time with 0.67 disengagements and 5.33 interventions per run [1]. The expectation was that video-based temporal models, with access to motion history, would improve on this. They did not. VideoMAE, VideoMamba, and two sequential variants all performed worse in real-world deployment despite comparable offline training metrics. The Sequence ViM model was tested once and abandoned on safety grounds.

The failure is architectural rather than computational. All models ran above the required frame rate; the problem lies in how their temporal representations are constructed. Video models designed for classifying pre-recorded footage aggregate information across their entire input window before producing an output. Applied to real-time control, this produces perceptual lag: the model’s output reflects a blended summary of the input history rather than the current scene, at exactly the moment when the most recent observation is most relevant to the control decision. Several design choices contribute to this. Bidirectional scanning borrows from frames that do not yet exist at inference time.

A fixed input window of eight frames accumulates nearly half a second of history regardless of how much has changed. Frame spacing and temporal weighting are treated as constants, indifferent to vehicle speed or scene dynamics.

Quirke-Brown’s analysis of the sequential models identifies lag directly as the operational failure mode: delayed steering responses at consistent locations recur across runs and are attributed to output lag rather than path deviation [1]. Causal forward-only scanning, reduced input windows, and adaptive temporal fusion are explicitly recommended as next steps.

This thesis investigates those recommendations on the UWA campus route. A new dataset is collected on campus and VMamba is retrained on it to establish a comparable baseline in this environment. Four architectural modifications are then implemented and evaluated: replacing bidirectional with forward-only causal SSM scanning; reducing and adaptively controlling the input frame window; introducing an explicit recency prior into the temporal weighting mechanism; and varying inter-frame spacing. Each modification targets a distinct contributor to perceptual lag. All model variants are screened in EyeSim simulation to verify training viability before any on-road testing, consistent with the pipeline from prior work [1]. The hypothesis is not that temporal context is harmful, Quirke-Brown’s parking and reversing results show it can help. But the way temporal information is aggregated for real-time control is what needs to change.

2 Literature Review

Quirke-Brown [1] documented a consistent deployment failure on the nUWay shuttle: every temporal video model underperformed a single-frame VMamba [3] baseline in real-world continuous driving. The root cause is bidirectional frame aggregation: when a model scans both forward and backward, the representation of frame t is conditioned on frames $t+1, t+2, \dots$ that do not exist at inference time. The model predicts from a representation calibrated for richer input than it receives, producing a systematic delay in response to the current scene.

2.1 State Space Model Foundations

Modern State Space Models trace their form to the HiPPO framework [4], which initialises the state transition matrix A by projecting continuous-time input onto an optimal polynomial basis. The result is a compressed historical representation maintained in constant time per step. S4 [5] operationalised HiPPO into a practically trainable architecture through normal-plus-low-rank structure and Cauchy kernel representations, with competitive results on the Long Range Arena benchmark. The simplified S5 architecture [6] demonstrated that structured SSMs scale linearly in sequence length, a direct contrast with the quadratic complexity of self-attention. Both S4 and S5 are Linear Time-Invariant: their transition parameters are fixed constants that cannot filter inputs based on content.

Mamba [7] addresses this by making the discretisation timestep Δ , the input matrix B , and the output matrix C functions of the current input. The parameter Δ governs the balance between absorbing the current token and preserving accumulated state: large values absorb the current token while discarding prior state; small values preserve state and attenuate the current input. This input-dependent gating, combined with a hardware-aware parallel scan, yields Transformer-quality performance at linear complexity. Mamba-2 [8] formalises a structural duality between SSMs and structured attention; this theoretical link underpins the MTIL architecture discussed in Section 2.7.

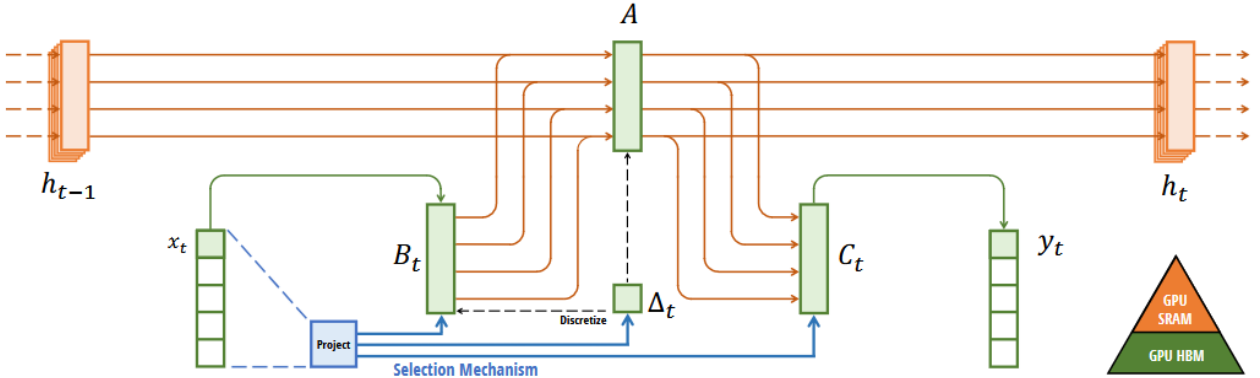


Figure 2: The Mamba selective SSM. Input x_t is projected to produce input-dependent parameters B_t , C_t , and discretisation timestep Δ_t , which modulate the fixed state transition matrix A . Adapted from [7].

For real-time deployment, Mamba supports a causal recurrent mode that processes one step at a time against a fixed-size hidden state, the architectural basis for Modification 1. Because Δ is already input-dependent, Mamba can in principle learn appropriate temporal emphasis without explicit intervention. Modification 3 goes further by imposing a recency prior on Δ , motivated by control-theoretic

arguments about observation staleness in reactive driving; whether this improves on learned weighting is an empirical question this study addresses.

2.2 SSM Adaptation to Visual Data

Self-attention [9] computes pairwise interactions between every input element at quadratic cost. Dosovitskiy et al. [10] applied this to images via fixed-size patch tokens; Swin [11] reduced cost with local shifted windows at the expense of a limited global receptive field.

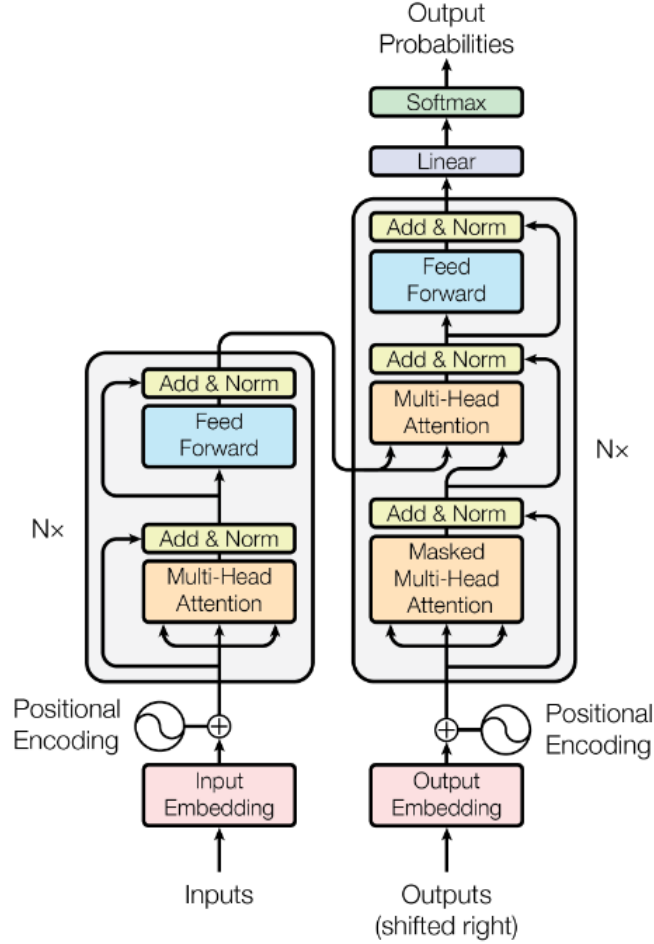


Figure 3: The Transformer model architecture, Adapted from [9].

Vision Mamba [12] resolves the mismatch between Mamba’s 1D recurrence and unordered image patches by scanning forward and backward and merging outputs so each patch draws on the full image. Removing the backward scan costs 7 top-1 accuracy points on ImageNet and 3.6 mIoU on semantic segmentation. This is a concrete cost of removing bidirectionality in the spatial domain, but the spatial and temporal cases are structurally different: future frames do not exist at inference, so scanning them is architecturally incompatible with causal deployment.

VMamba [3] operates on single frames and avoids the temporal problem entirely. The 2D Selective Scan (SS2D) module traverses each image along four spatial paths and gives each token a global receptive field without crossing temporal boundaries. In deployment on the nUWay shuttle, VMamba achieved 98.23% autonomous driving time with zero disengagements in its best run [1]. Video-

Mamba [13] extends VMamba to video with bidirectional 3D spatiotemporal scans, preserving SSM linear complexity while enabling long-range temporal modelling. It is the base architecture for this study’s four modifications because it directly extends VMamba to the temporal domain, inheriting the same SS2D spatial backbone while adding the temporal dimension where the deployment failure occurs. Modifying VideoMamba therefore isolates the temporal aggregation mechanism as the independent variable, holding spatial processing constant relative to the VMamba baseline.

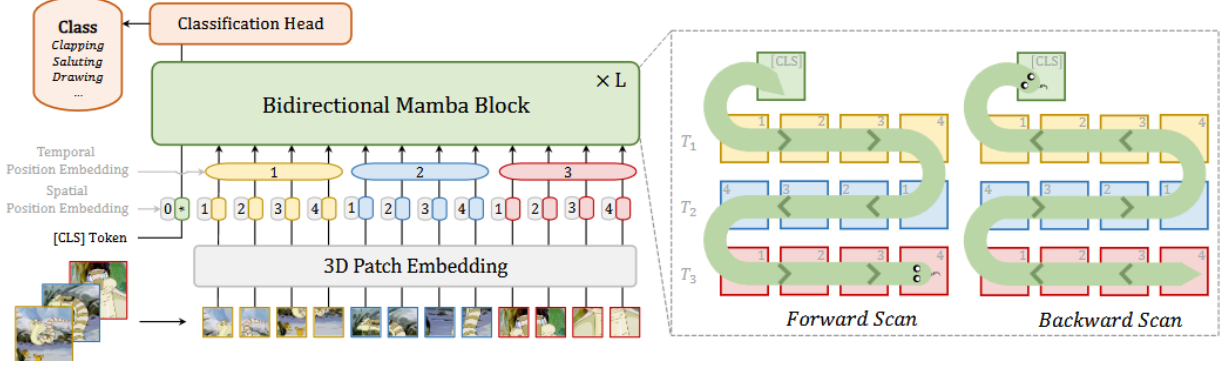
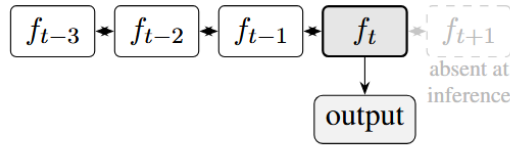


Figure 4: VideoMamba architecture. Video frames are tokenised via 3D patch embedding and processed through L stacked Bidirectional Mamba Blocks, with both forward and backward spatiotemporal scans aggregating context across patches and timesteps. Adapted from [13].

2.3 Temporal Video Models and the Deployment Gap

The models tested in [1] span the main families of temporal video modelling. TimeSformer [14] applies joint space-time self-attention across all spatiotemporal tokens, attending symmetrically to past and future. ViViT [15] factorises this into separate spatial and temporal attention components; temporal attention remains bidirectional. ViS4mer [16] pairs a transformer spatial encoder with an S4 temporal decoder; S4’s recurrence is causal, but as an LTI model it cannot selectively gate content across the temporal sequence. VideoMAE [17] uses masked autoencoder pre-training with 90–95% tube masking to reach strong action recognition results from small datasets.

(a) Bidirectional scan (all models tested in [1])



(b) Forward-only (causal) scan (Modification 1)

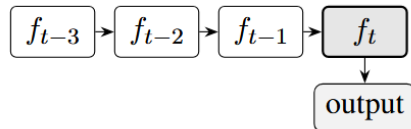


Figure 5: Bidirectional versus forward-only causal scanning. In (a), each frame’s representation is conditioned on f_{t+1} , which does not exist at inference time; the output blends past and a future the model was trained on but cannot observe during deployment. In (b), the forward-only scan aggregates only past and present frames, producing an output that reflects the current scene without dependency on unavailable future context.

Adjusted real-world deployment performance across all models is summarised in Table 1. VMamba led on every metric that matters for continuous driving. The failures were more concrete than statistical overlap might suggest: VideoMAE reached 80.60% autonomous time with 22 interventions and 7 disengagements; VideoMamba averaged $89.05 \pm 4.46\%$ with a mean of 8.67 interventions and 5.67 disengagements; Sequence ViM reached 95.07% but with 13 disengagements, a model that repeatedly lost control. Sequence VMamba came closest at $94.79 \pm 4.33\%$, overlapping VMamba’s interval, though the mean difference remained below statistical significance [1].

Table 1: Adjusted real-world deployment performance across temporal and single-frame models evaluated on the Eglinton suburban route. Results exclude disengagements and interventions in untrained or out-of-distribution scenarios. Mean $\pm$ standard deviation where multiple runs were conducted; single-run models have no standard deviation reported. Reproduced from [1].

Metric	VideoMAE	VideoMamba	Seq. ViM	Seq. VMamba	VMamba
Drive time (min)	21.72	20.76 ± 1.02	23.29	22.83 ± 0.64	18.42 ± 1.46
Autonomy (%)	80.60	89.05 ± 4.46	95.07	94.79 ± 4.33	95.77 ± 2.73
Disengagements	7	5.67 ± 4.16	13	7.00 ± 2.00	0.67 ± 0.58
Interventions	22	8.67 ± 3.51	7	6.33 ± 0.57	5.33 ± 1.53

The deployment data is not uniformly negative for temporal modelling. On low-speed manoeuvring tasks, Sequence VMamba produced the highest parking result of any model (2.88/4.00 against VMamba’s 2.58/4.00), and VideoMAE achieved perfect reverse scores (2.00/2.00) [1]. Quirke-Brown attributes this to temporal context providing genuine value for position-dependent low-speed control. The continuous driving failure is not evidence that temporal context is useless, instead it is evidence that bidirectional aggregation prevents the model from using it cleanly under causal constraints. Quirke-Brown explicitly recommends forward-scanning Mamba blocks as a direction for future work.

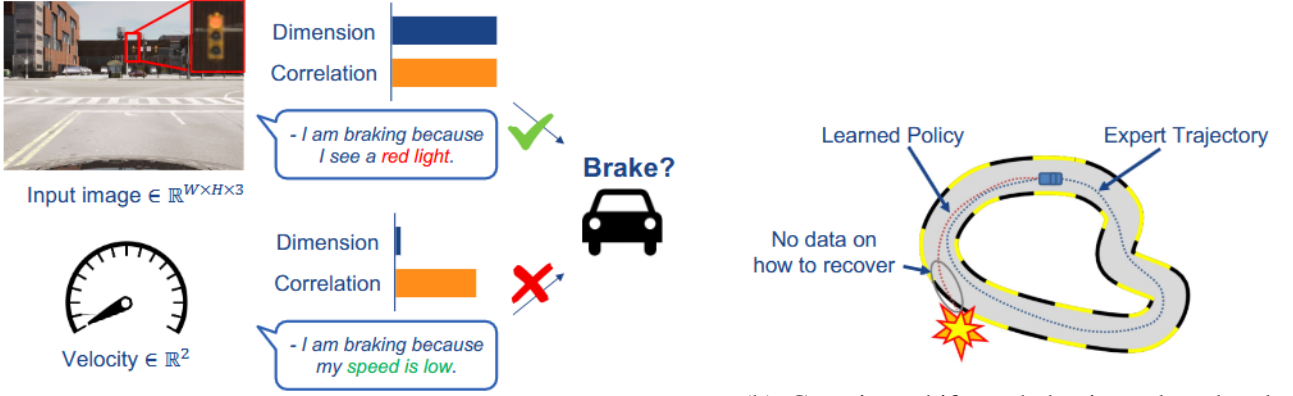
2.4 End-to-End Imitation Learning and the Deployment Gap

The nUWay approach traces to Bojarski et al. [18], demonstrated that a CNN trained on human driving data could control a vehicle in real traffic before being superseded by transformer and SSM architectures.

Ross et al. [19] established why open-loop imitation learning fails in closed-loop deployment. Behaviour cloning treats each state-action pair as i.i.d.; at deployment, the policy’s own actions generate out-of-distribution states and small errors compound. This implies that each additional historical frame adds a compounding source of distributional shift, since that frame was encountered under a different policy state, a second motivation for Modification 2, independent of the lag argument: a shorter temporal window reduces both aggregation delay and compounding distributional shift.

Chen et al. [20] survey end-to-end autonomous driving at scale, identifying covariate shift, causal confusion, and the open/closed-loop evaluation gap as the field’s central unresolved problems, with the gap worsening at longer temporal windows. Tampuu et al. [21] find the same pattern across architectures: simulation and offline performance are systematically poor predictors of real-world deployment success. Both surveys corroborate the nUWay result as an instance of a wider phenomenon

rather than an anomaly of the specific architecture tested.



(a) Causal confusion: end-to-end models may latch onto low-dimensional spurious correlates of the correct action rather than the true causal feature.

(b) Covariate shift: a behaviour-cloned policy drifts off the expert trajectory into states unseen during training, with no recovery data available.

Figure 6: Failure modes of open-loop imitation learning in closed-loop deployment. Adapted from [20].

2.5 Frame Sampling and Temporal Window Design

Window length and frame spacing are independent design parameters. Modification 2 addresses window length; Modification 4 addresses spacing.

Wang et al. [22] established uniform sparse sampling as the standard with Temporal Segment Networks: the clip is divided into equal temporal segments and one frame drawn from each. This applies directly to the 8-frame uniform input window used by all temporal models in [1]. SlowFast [23] challenged single-rate sampling with two parallel pathways at different frame rates, showing that temporal resolution encodes qualitatively different information at different rates. This motivates the hypothesis that optimal frame spacing for real-time driving may vary with vehicle speed, since speed determines how quickly the visual scene changes.

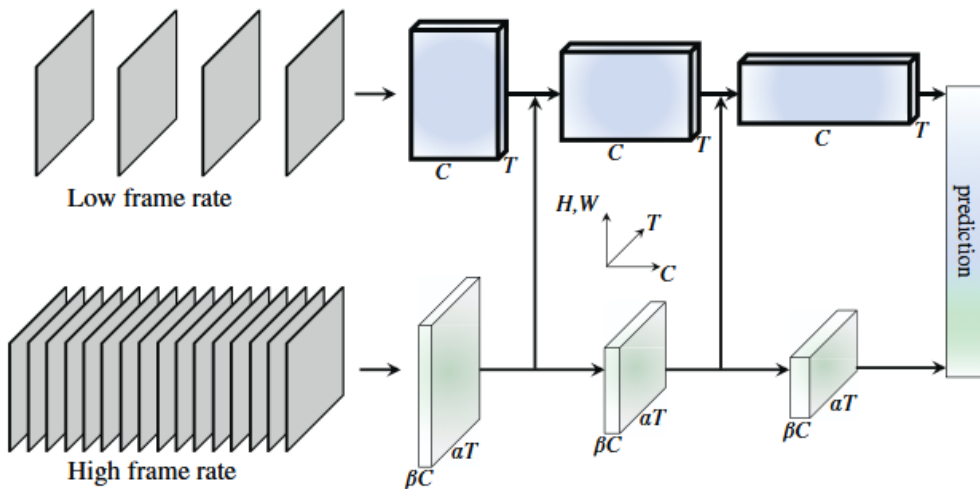


Figure 7: The SlowFast network architecture: a low-frame-rate *Slow* pathway captures spatial semantics whilst a high-frame-rate *Fast* pathway captures fine-grained motion. Lateral connections fuse both representations before prediction. Adapted from [23].

Zhi et al. [24] demonstrated with MGSampler that motion-guided adaptive frame selection consistently outperforms uniform sampling: dense static segments contribute little while high-motion moments carry disproportionate discriminative content. A causal implementation for real-time driving must compute frame selection using only past information; this study investigates fixed spacing variants as a tractable first step toward speed-conditioned adaptive sampling.

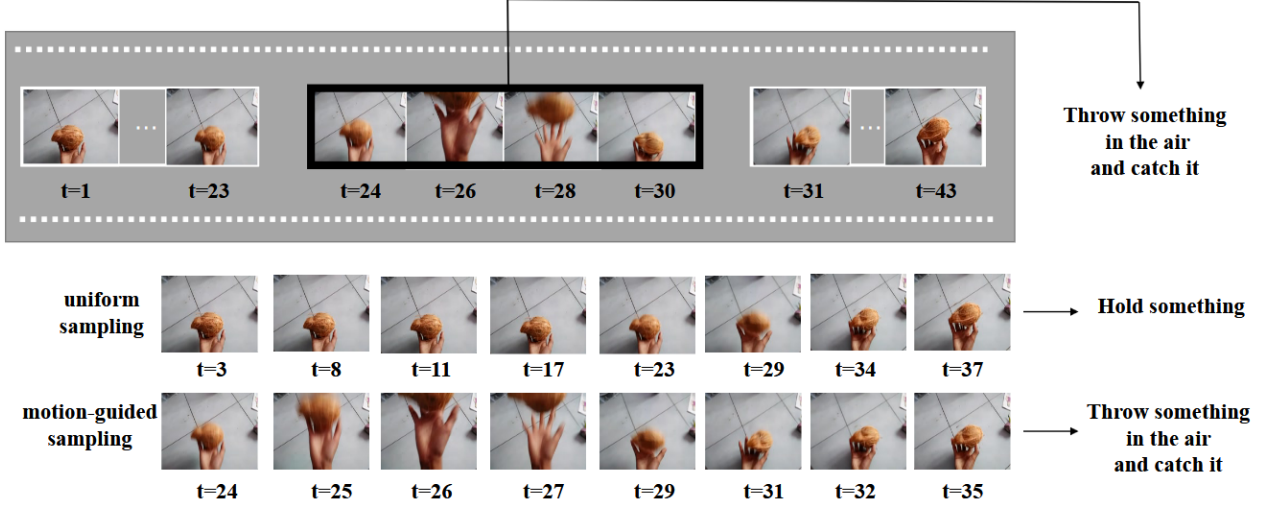


Figure 8: Uniform sampling (middle) distributes frames evenly across a clip and misses the peak-motion window, yielding an incorrect prediction. Motion-guided sampling (bottom) concentrates frames on high-motion moments and recovers the correct label. This motivates Modification 4: fixed but variable temporal spacing as a tractable first step toward speed-conditioned adaptive sampling in real-time driving. Adapted from [24].

2.6 Adaptive Temporal Weighting

Huang et al. [25] challenged the temporal-invariance assumption of standard convolutional video models with TAdaConv, generating adaptive per-frame calibration weights α_t from each frame’s local and global temporal context. TAdaConv establishes that different frames should receive different weights; it does not establish that newer frames should receive higher weights, as its weights are content-adaptive and generated with bidirectional context rather than encoding a recency prior.

The recency prior in Modification 3 rests on a control-theoretic argument: a frame from k steps ago describes a scene the vehicle has since left, so relevance declines with k . The mechanism applies a monotonically decreasing position-dependent offset to the input-projected Δ values before discretisation. In a forward scan, smaller Δ at early steps retains accumulated state while weakly absorbing distant inputs; larger Δ at recent steps absorbs current inputs while discarding stale state, biasing the hidden state toward recent frames without altering the parallel training scan.

Wang et al. [26] provide the strongest counterargument, that SSMs already exhibit inherent recency bias that impairs recall of distant information. Their concern targets long-context language tasks; the nUWay setting is a short driving window where the failure mode runs in the opposite direction. Whether amplifying recency in this short window produces the targeted improvement in reactive control is what Modification 3 tests.

2.7 Causal Sequence Modelling in Robotic Control

The closest empirical evidence that causal SSM processing works in deployed real-time control comes from the robotics literature.

Zhou et al. [27] introduce MTIL, which uses Mamba’s [7] recurrent step function to process robot observations one timestep at a time in strict causal order. Validated on simulated benchmarks and real dual-arm manipulation tasks, MTIL substantially outperforms the Action Chunking Transformer and Diffusion Policy on tasks requiring long-horizon temporal reasoning. Jia et al. [28] introduce MaIL, using Mamba as the backbone for behaviour cloning and diffusion-based imitation policies. MaIL consistently outperforms Transformer-based policies with limited data on the LIBERO benchmark, with the advantage attributed to Mamba’s selective state retention. Across both papers, Mamba-based architectures are competitive with or superior to Transformers in the imitation learning setting which is direct empirical precedent for Modification 1 in robotic control.

One MTIL result bears on Modification 2: a short-history (10-step) variant performs similarly to Transformers, while full-history MTIL dominates both. MTIL’s manipulation tasks require recall of completed sub-tasks whose context may lie arbitrarily far in the past; reactive driving is different, where the failure mode is perceptual lag from short-range aggregation rather than missing long-range memory. MTIL’s finding is a genuine empirical tension for Modification 2, but its structural basis does not transfer directly.

No existing work directly ablates bidirectional versus causal Mamba within the same architecture on the same task. MTIL operates causally by design; the vision Mamba architectures in Section 2.2 are bidirectional by design and not evaluated in a causal configuration. That comparison, on a real deployed vehicle, is what the present study contributes.

Table 2: Literature mapped to each modification. *Supporting* entries provide direct motivation; *Opposing* entries represent empirical or theoretical tension, not absence of evidence; the *Rebuttal* column explains why the tension does not preclude investigation.

Modification	Supporting	Opposing	Rebuttal
Mod 1: Causal-only scanning	[1]: bidirectional lag is the identified cause of deployment failure; forward scanning explicitly recommended. [27]: causal Mamba step-function outperforms Action Chunking Transformer on deployed real robot. [28]: Mamba-based IL competitive with Transformers under limited data.	[12]: removing the backward spatial scan costs 7% top-1 on ImageNet and 3.6 mIoU on segmentation.	The cost is measured in the <i>spatial</i> domain where backward tokens exist at inference. Future <i>temporal</i> frames do not exist at deployment; backward temporal scan is architecturally incompatible with causal inference, making the performance loss irrelevant to this setting.
Mod 2: Reduced frame win- dowing	[19]: compounding distributional shift grows with history length in behaviour cloning. [20]: open/closed-loop performance gap widens at longer temporal windows. [1]: continuous-driving failure attributed partly to long-window aggregation.	[27]: full-history MTIL dominates a 10-step short-history variant. [1]: temporal context aids low-speed manoeuvring.	MTIL requires recall of arbitrarily distant sub-task completions; reactive driving fails from short-range perceptual lag, not missing long-horizon memory. The failure modes are structurally opposite. The low-speed benefit does not contradict window reduction for high-speed continuous driving.
Mod 3: Recency- biased Δ	[7]: Δ is already input-dependent; Mod 3 adds a position-monotonic prior on top of learned weighting. Control-theoretic: a frame k steps old describes a scene the vehicle has since left; relevance decays with k .	[26]: SSMs exhibit inherent recency bias that already impairs recall of distant tokens. [25]: TAdaConv shows adaptive weights are content-driven, not position-driven.	The recency bottleneck concern is documented in <i>long-context language</i> tasks; the nUWAy window is short and the failure mode runs in the opposite direction. TAdaConv supports differential per-frame weighting but uses bidirectional context, which is incompatible with causal deployment; the objection is to the mechanism, not the principle.
Mod 4: Variable frame spacing	[23]: multi-rate sampling encodes qualitatively different temporal information at different rates. [24]: non-uniform motion-guided sampling outperforms uniform sampling on action recognition.	[22]: uniform sparse segment sampling is the dominant established baseline for video understanding.	TSN targets recognition from pre-recorded clips where scene change rate is fixed. Real-time driving requires spacing calibrated to vehicle speed, since speed determines how fast the visual scene changes; uniform spacing is a suboptimal default for this regime rather than an established optimum.

3 Project Objectives

3.1 Research Hypothesis

The hypothesis is that targeted modifications to VideoMamba’s temporal aggregation mechanism without replacing the architecture completely, can eliminate perceptual lag sufficiently for the model to match or exceed VMamba’s autonomous driving performance. At least one of the four proposed configurations will achieve autonomous driving time and disengagement counts within the range of VMamba’s best deployment results.

3.2 Research Objectives

Objective O0 establishes a campus-specific VMamba baseline. Quirke-Brown’s work was conducted on the Eglinton suburban route; the campus environment has different geometry, lighting, and manoeuvre types, so a dedicated retraining step is required to isolate route-specific effects from architectural differences. Objectives O1-O4 each target a distinct contributor to perceptual lag, with results compared directly against that baseline.

- O0. Establish a campus-specific VMamba baseline.** Collect a new driving dataset on the UWA campus route and retrain VMamba under matched conditions to Quirke-Brown’s best-performing configuration [1]. All subsequent architectural comparisons are benchmarked against this model.
- O1. Evaluate the effect of forward-only causal scanning on deployment performance.** Implement a VideoMamba variant in which the bidirectional temporal SSM scan is replaced by a single forward pass using Mamba’s native causal recurrent mode. Screen offline against the campus baseline on per-behaviour validation accuracy, then evaluate in real-world deployment on the shuttle using Quirke-Brown’s protocol.
- O2. Characterise the effect of temporal window size and adaptive windowing on deployment performance.** Implement and evaluate VideoMamba variants with reduced fixed-length input windows and, as a candidate extension, a speed-conditioned adaptive window that adjusts frame count based on vehicle velocity. Assess the trade-off between aggregation delay and available temporal context across offline and real-world metrics.
- O3. Evaluate the effect of an explicit recency prior on temporal aggregation.** Implement a position-dependent bias on Mamba’s Δ parameter that monotonically increases toward more recent frames, causing the hidden state to weight recent input more heavily. Assess whether this improves reactive control in deployment, and evaluate against the counterargument from Wang et al. [26] that amplifying recency in SSMs impairs information retention.
- O4. Characterise the effect of inter-frame spacing on deployment performance.** Implement VideoMamba variants with fixed temporal intervals between sampled frames, spanning dense to sparse sampling, and identify whether any spacing regime improves real-world driving performance relative to the uniform baseline. Evaluate a speed-conditioned variant if fixed-spacing results indicate a meaningful effect.

Table 3: Summary of the four VideoMamba modifications. Each objective targets a distinct architectural contributor to perceptual lag.

Obj.	Modification	Mechanism targeted	Key implementation change
O1	Forward-only causal scanning	Bidirectional SSM temporal scan	Replace backward scan with Mamba native forward recurrent step; no future frames used
O2	Adaptive frame windowing	Fixed 8-frame input window	Reduce fixed window length; speed-conditioned adaptive sizing as candidate extension
O3	Recency prior on Δ	Uniform temporal weighting	Monotonically increasing position-dependent offset applied to Δ before discretisation
O4	Variable inter-frame spacing	Fixed uniform inter-frame intervals	Fixed-stride variants spanning dense to sparse; speed-conditioned spacing as candidate extension

3.3 Significance

If any modified variant matches VMamba’s deployment performance, it would show that the gap between offline-trained video models and single-frame SSMs is a fixable architectural artefact rather than a fundamental property of temporal modelling. The nUWay project would then have a route to a model that can use motion history unlike VMamba. Quirke-Brown’s parking and reversing results [1] show that temporal context is useful when the aggregation mechanism does not obstruct it.

The research contribution is a within-architecture ablation that prior work has not run: bidirectional versus causal temporal Mamba scanning on a real deployed vehicle, with a matched evaluation protocol. Vision Mamba [12] ablates bidirectionality in the spatial domain; MTIL [27] and MaIL [28] use causal Mamba recurrence in robotic manipulation; neither tests this trade-off in a driving context on a real vehicle.

4 Project Process

The project runs two parallel workstreams that converge at real-world evaluation. The first uses EyeSim simulation to verify that proposed architectural modifications produce viable training behaviour before any on-road testing. The second is continuous campus data collection on the nUWay 4 shuttle throughout the semester. Once simulation experiments confirm modification viability, the final models are trained on the accumulated campus dataset and evaluated in deployment.

4.1 Simulation Environment

EyeSim [29] was selected over CARLA [30] because the two simulators use incompatible control interfaces. CARLA models vehicle dynamics through acceleration commands, whereas the nUWay shuttle operates on direct speed and steering control. EyeSim uses the same interface as the shuttle hardware, so simulation and campus training data share the same label format without conversion.

The EyeSim environment uses a virtual Carolo Cup track [1] with an S4 Eyebot robot as the data collection platform, as shown in Figure 9. Data is collected by manually driving the track, with images and corresponding speed and steering commands recorded as NumPy array files. The target dataset size is approximately 100,000 samples, consistent with the volume Quirke-Brown found sufficient for stable model convergence [1].



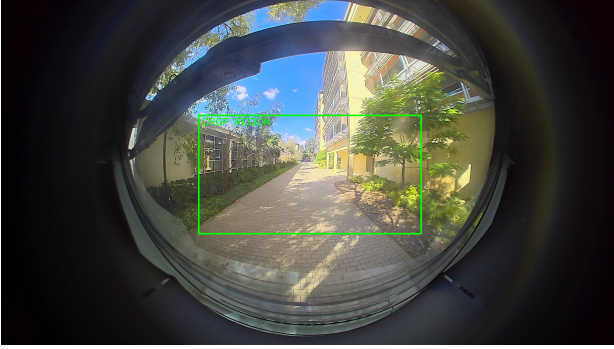
Figure 9: The virtual Carolo Cup track layout in EyeSim, navigated by an S4 Eyebot robot during initial simulation screening. All model variants complete this track before any on-road testing is attempted. Figure reproduced from [31].

4.2 Campus Data Collection

Real-world data collection is conducted on the UWA campus route using the nUWay 4 shuttle. A ROS2 preprocessing node crops each frame to the road-dominant region and resizes to 180×400 pixels, publishing both greyscale and RGB outputs at 20 Hz to match Quirke-Brown’s acquisition

protocol [1]. Both image channels and the corresponding manual speed and steering commands are recorded simultaneously, removing the need for separate collection runs per channel.

Sessions are conducted at different times of day and under varying lighting conditions to build distributional variety. Speed values are normalised to $[0, 1]$ and steering to $[-1, 1]$, consistent with Quirke-Brown’s normalisation scheme [1]. The data collector drives in a consistent, smooth style throughout each session, as the expert demonstrator in the imitation learning framework [18].



(a) Raw camera frame (full resolution).



(b) Preprocessed frame: road-dominant crop, resized to 180×400 px, grayscale.

Figure 10: Effect of the ROS2 preprocessing node on a sample campus frame. The crop removes the sky and vehicle bonnet, retaining the road-dominant region relevant to speed and steering prediction.

4.3 Model Configurations

All models are implemented in PyTorch and trained on the UWA Robotics Lab Titan computing cluster. Two model configurations form the basis of the comparison.

The first is a VMamba model configured identically to the best-performing baseline reported by Quirke-Brown [1], retrained on campus data to establish a route-specific baseline (Objective O0). The second is a modified VideoMamba model [13] with the temporal aggregation mechanism altered according to each of the four proposed modifications: forward-only causal scanning (Objective O1), reduced and adaptive frame windowing (Objective O2), an explicit recency prior on Δ (Objective O3), and variable inter-frame spacing (Objective O4). Each modification produces a distinct variant trained and evaluated against the VMamba baseline under matched conditions.

Greyscale and RGB variants of each configuration are evaluated as a secondary comparison, using the same recorded sessions with no additional driving time required.

4.4 Training Procedure

All training follows Quirke-Brown’s established procedure [1]. Loss is mean absolute error over speed and steering outputs; MAE prevents the larger-magnitude speed values from dominating back-propagation in a way that MSE would. An adaptive learning rate schedule decays on validation loss plateaus. Data augmentation includes horizontal image shifting with proportional steering label adjustment to generate recovery-from-edge examples, and synthetic motion blur to match real-world camera conditions. The training set is split into per-behaviour subsets: lane following, pull-in, and

reverse. Model accuracy is assessed by the proportion of predictions within 5% and 10% of ground-truth speed and steering, rather than loss alone.

4.5 Evaluation Protocol

Evaluation runs in two stages. All model variants are first assessed offline on a held-out validation subset; only variants meeting or exceeding the VMamba baseline on per-behaviour accuracy proceed to on-road testing, consistent with Quirke-Brown’s gating criterion [1].

Passing variants are deployed on the nUWY 4 shuttle for real-world evaluation. Performance is measured by autonomous driving time, disengagements per run, and path deviation from the route centreline. A minimum of three runs per variant accounts for the run-to-run variance documented in Quirke-Brown’s results [1]. The primary evaluation question is whether any modified VideoMamba variant achieves autonomous performance within the range of VMamba’s best deployment results.

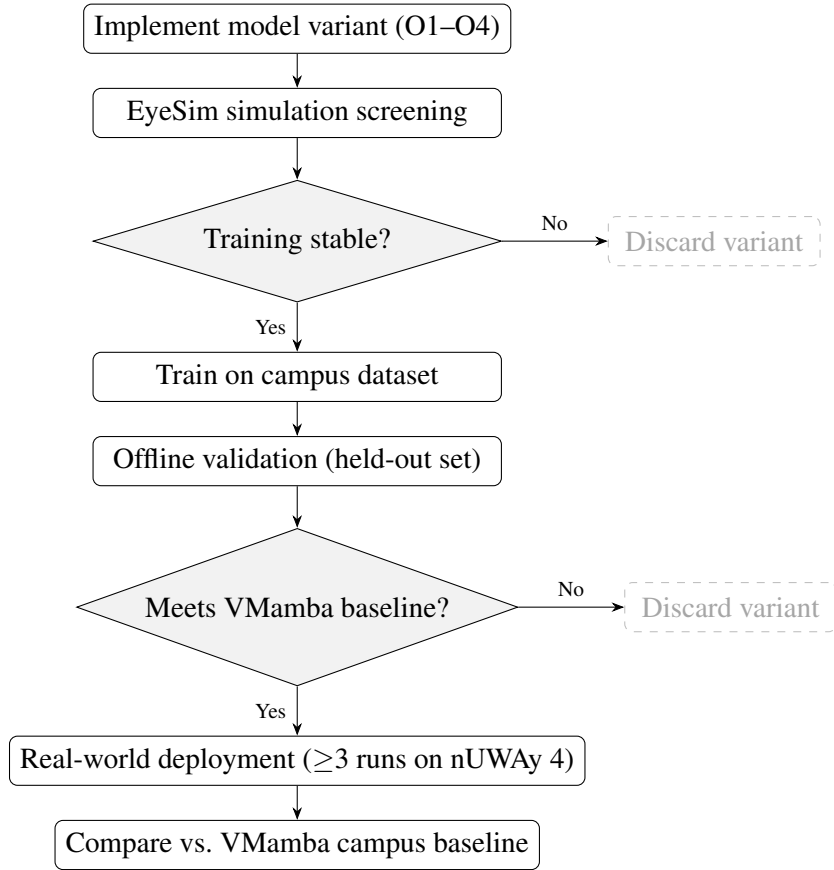


Figure 11: Evaluation pipeline applied to each model variant. EyeSim simulation screening filters unstable modifications before any campus training is attempted. Offline validation against the VMamba campus baseline gates real-world deployment, consistent with Quirke-Brown’s gating criterion [1].

5 Timeline and Risk Management

The project timeline is provided as a Gantt chart in Appendix A. The critical path runs through campus data collection, model training, and on-road evaluation in sequence: a sustained delay in data collection directly postpones all training, which in turn postpones deployment testing.

All Semester 1 milestones are on schedule and the Project Proposal milestone (Week 12) is on track for on-time submission. No tasks have slipped from the planned timeline.

Table 4 summarises the principal risks, their likelihood and consequence, and the management strategies in place. No significant environmental risk is associated with this project.

Table 4: Risk register for GENG4411 project.

Risk	Likelihood	Consequence	Mitigation
Injury during on-road testing	Low	High	Human supervisor present at all times; off-line screening gate prevents untested models from reaching deployment.
Campus route access disrupted (weather, events, scheduling)	Medium	Medium	Collection sessions scheduled early and spread across the semester; EyeSim-only evaluation is a valid fallback if campus access is lost.
Architectural modification fails to train stably	Medium	Medium	EyeSim screening filters unstable variants before campus training begins; modifications are implemented incrementally with unit tests on subcomponents.
Cluster unavailability delays training	Low	Medium	Training jobs queued in advance; VMamba baseline trained first as highest-priority.
Data loss (hardware failure or accidental deletion)	Low	High	All datasets and checkpoints backed up to UWA network storage after each session.

6 Progress to Date

Progress is on schedule against the Gantt chart. Completed tasks are as follows.

The literature review is complete. The project repository has been organised; the training scripts of primary interest: VMamba and VideoMamba variants for both EyeSim and the nUWAr shuttle; have been identified and reviewed in detail.

The EyeSim simulation environment is configured with the Carolo Cup track. The data collection pipeline for simulation is currently being developed; model training in EyeSim will follow once collection is operational, consistent with the gating criterion in Figure 11.

Rosbag recordings have been collected on the campus route. A preprocessing node was validated against the collected rosbags. Crop parameters were tuned visually by running the node on exist-

ing recordings and inspecting output frames; the retained region removes the sky and vehicle bonnet across tested lighting conditions, retaining the road-dominant area as shown in Figure 10. Both greyscale and RGB outputs were verified at 20 Hz with synchronised `TwistStamped` labels. Table 5 summarises the validated pipeline parameters.

Table 5: Validated ROS2 preprocessing pipeline parameters.

Parameter	Value
Input topic	<code>/CameraFront/image_raw</code>
Crop removed (top / bottom)	34% / 32% of frame height
Crop removed (left / right)	32% / 32% of frame width
Output resolution	400×180 px
Output rate	20 Hz (wall-clock throttle)
Greyscale storage rate	≈ 1.4 MB/s
RGB storage rate	≈ 4.1 MB/s
Combined storage rate	≈ 5.5 MB/s (≈ 330 MB/min)
Label synchronisation	<code>/cmd_vel</code> republished as <code>TwistStamped</code>

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A Gantt Chart

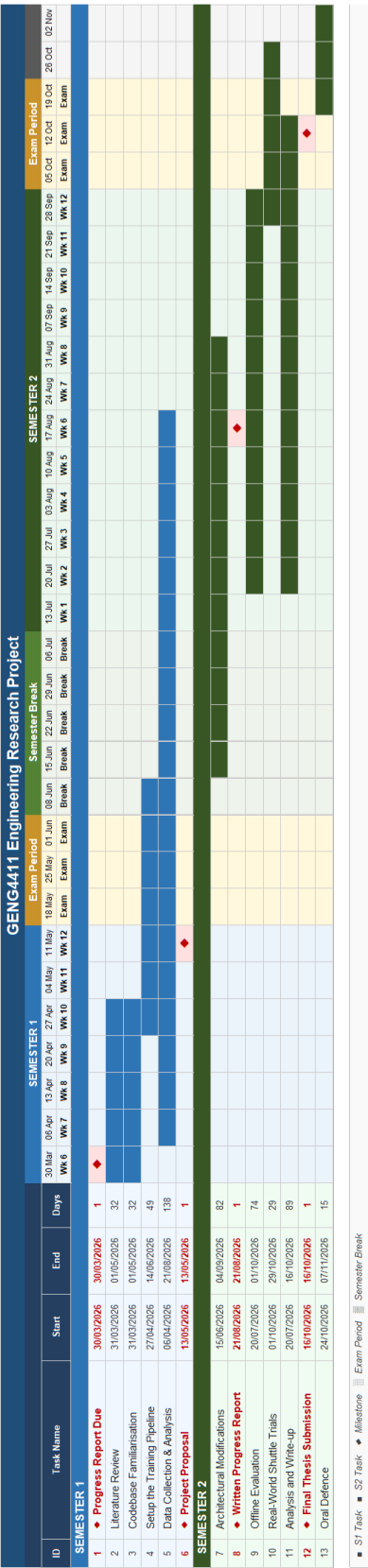


Figure 12: Gantt chart showing tasks and milestones across Semester 1 and Semester 2.